ATTRACTOR Special Session

Toward Justifiable Trust in Autonomous Systems

Trust, Trustworthiness and Risk in Autonomous Decision Making

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Outline

- ATTTACTOR: a brief introduction
- Components of trust
- A component: XAI
- A component: predicting phase transitions
- Concluding remarks

ATTRACTOR Overview

Autonomy Teaming & TRAjectories for Complex Trusted Operational Reliability

- Team: Approx. 50 researchers and developers at LaRC, ARC, AFRC
- Student interns
- External collaborators: CMU, MIT, Tufts, UIUC, LMCO/Sikorsky, OGA
- Activity in NASA/TACP/CAS (Convergent Aeronautics Solutions) Project
- Duration: 30 months
- Co-PIs: Alexandrov & Allen
- Aeronautic Barrier: Build a basis for certification of autonomous systems by developing quantifiable criteria for trustworthiness and justifiable trust in multi-agent cyber-physical-human team operations, in safety-critical and time-critical environments.

Context: Assured Multi-agent System for Mission Planning and Execution

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Trustworthiness:

- Attribute of the cyber-physical system
- Requires context-driven metrics
- Achieved through *analyzable trajectories* and behaviors, supported by effective *teaming*, *learning* and *training*
- *Requires persistent M&S*

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Trust:

- Attribute of the users/observers/ controllers of the system
- Explainable AI (XAI) is critical in providing *clarity in decisions* and forensics, establishing *justifiable* trust, and defining dynamic roles and responsibilities

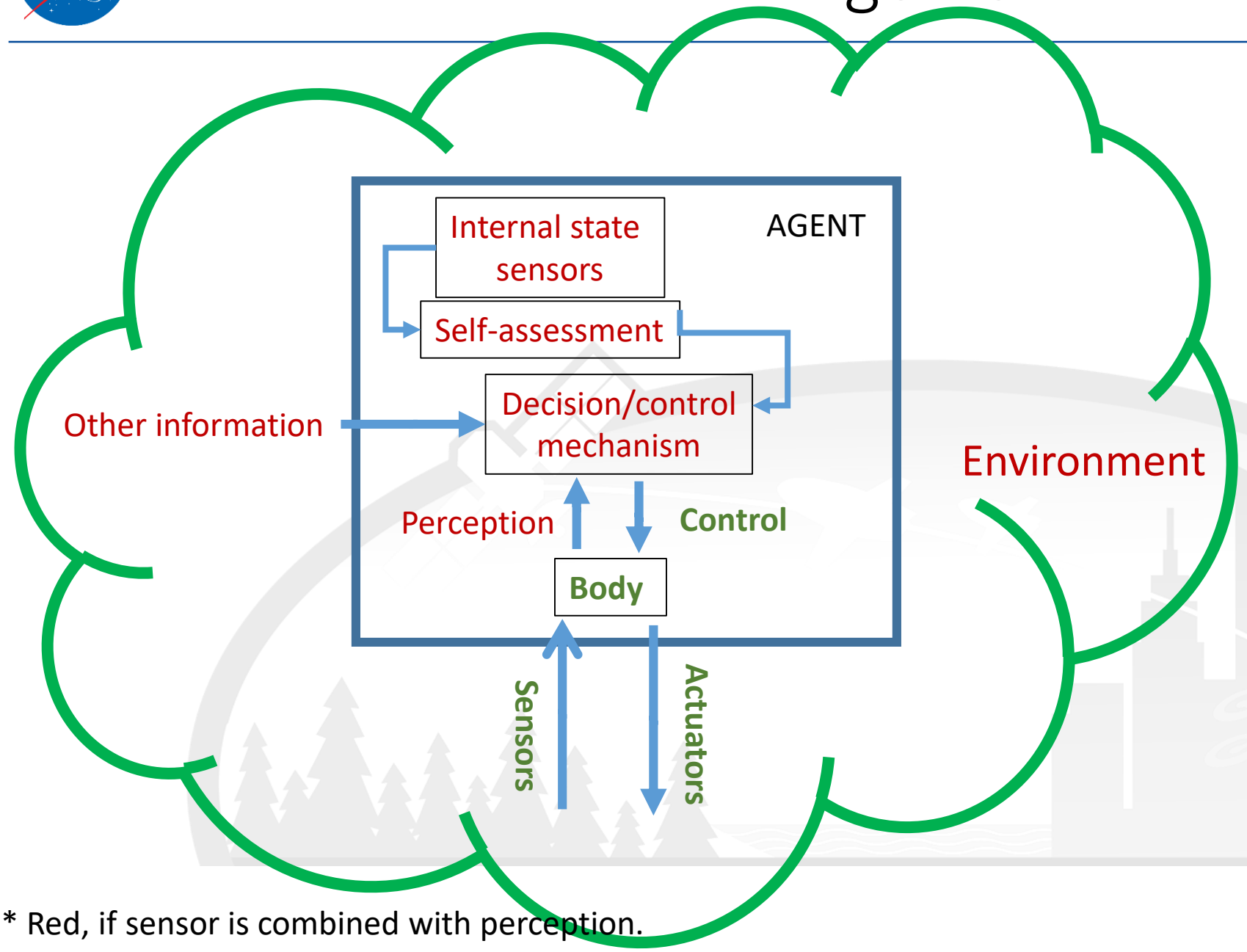
For human agents, natural interaction for mission planning and execution, supported by XAI, fosters trust and validates reliability

Justifiable Trust / Problem Setting

- Autonomy ≡ capability and authority to make decisions and act, independent of external control, in a specific environment, under a set of conditions
- Trust is based on the expectation of positive outcome of an action taken as a result of advice or direction from another agent
- Trust ≡ a state that permits one agent to act on advice or direction from another agent
  - Q: What features should characterize this state?
- Trust in multi-agent teams in safety-critical and time-critical environments must be developed during training



Agents



- trust ∧ trustworthiness ~ {reliability, efficiency, robustness, resilience, predictability of behavior, explicability, timely prediction of phase transitions, survivability} ~ 1/Risk
- Conjecture:

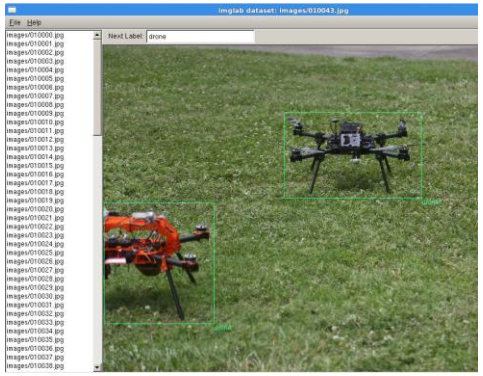
  Green: Amenable to traditional safety-based design

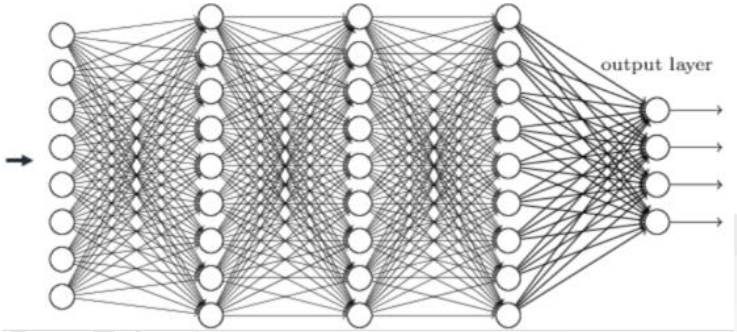
  Red: New risks; new approaches needed or bound complexity

* Red, if sensor is combined with perception.

Component: Explicability
("XAI"), H-M

- Regardless of algorithms, strong indication from experimental evidence that in H-M teams, explanation promotes productive H-M interactions (e.g., Lyons et al. 2016)





Two Quads →

XAI explanation: These are two UAVs. Because…. They are quadrotors. They have four propellers. They have four legs (landing gear). Etc. Or "I don't know".

- Examples of use cases for explicability
  - Computational decision making process during training of an expert and non-expert
  - Computational decision making process during operations
  - Visual recognition and classification
  - Forensics
- Working Hypothesis: Recipients of explanations would find others' explanations convincing if they fit within their own explicability framework (although the framework may require expansion)

XAI Questions (DARPA and Others)

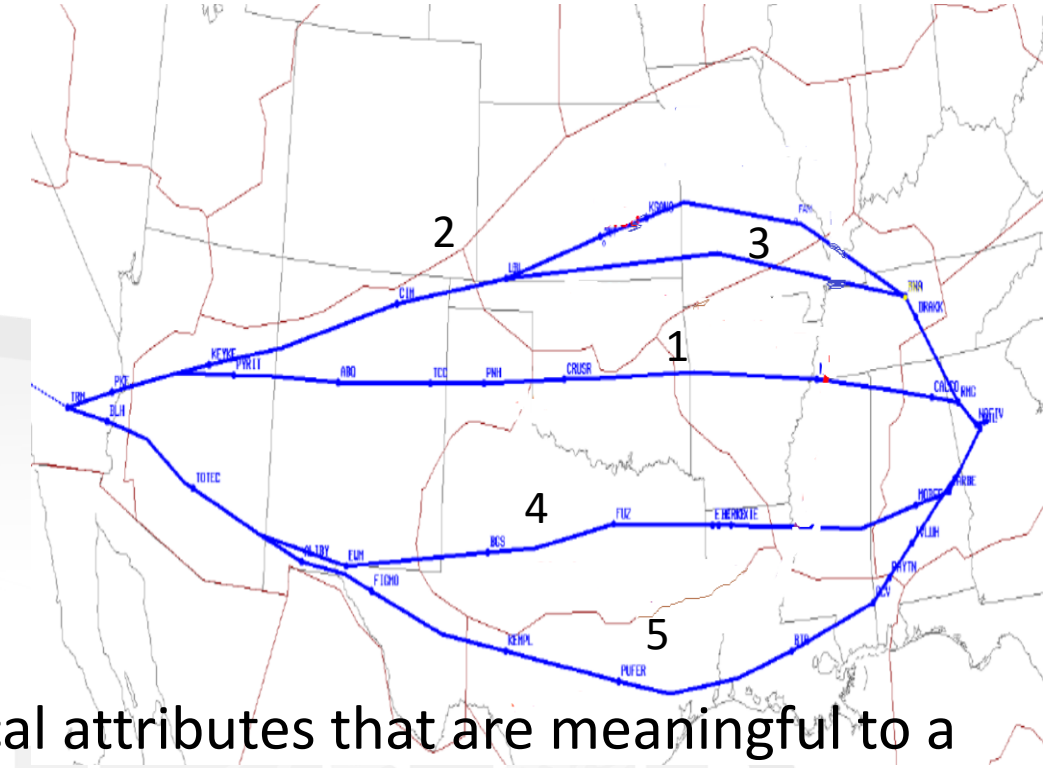
Unanswered Question	XAI	Comments
Why did you do that?	I understand why	Causal. May not be possible or meaningful in all cases.
Why not something else?	I understand why not	“Something else” is vague. Have to provide definition.
When do you succeed?	I know when you succeed/fail	Context/justification
When can I trust you?	I know when to trust you	Context /justification
How do I correct an error?	I know why you erred	Independent analysis needed / self-reflection

- Transparency: understand structure and functioning of the algorithms involved in decision-making.
- A posteriori explicability: ignore algorithmic structure and look for explanations that can be derived from the relation between input and output data

Proposed Approach to Explicability of Decision Making

- Consider explanations of a set of proposed trajectories:
  - Planned trajectory 1 is infeasible because violent weather is in the path of the old trajectory
  - Trajectories 2-5 are flyable and conflict-free
  - Trajectory 3 saves more fuel but flies through denser traffic that trajectory 2
  - …
- Common features of all acceptable explanations:
  - The old trajectory violated constraints
  - New trajectories are flyable and conflict-free (satisfy constraints); objectives have better values

Image courtesy FAA. Used for notional depiction of trajectories.



- Regardless of the algorithm, the trajectory has physical attributes that are meaningful to a human decision maker in terms of value functions associated with moving from point A to point B.
  - E.g., minimizing fuel expenditure, minimizing delays in reaching destination, maintaining safe distance from other aircraft, objects, and weather, etc.

Context and Goal Driven Explanation

- Convincing explanation must contain information about
  - Constraint violations and comparative values of the objectives and constraints between the current and proposed decision, such as a trajectory
  - Sensitivity (robustness information)
  - Overall risk information (uncertainty horizons, consequences implied or explicit)
  - Information about alternative decisions via what-if scenarios if sims are available
- Note: In M-M interactions, resolving information constraints also requires physical “explanation”, appropriately represented.
- (Approaches to image classification explicability will be discussed in later talks.)

Common Cognitive Model for Explanation
(Working Hypothesis)

- Regardless of the decision making algorithm, local (reward) and global (utility) objectives must be interpretable to H in M-H interactions
- Proposed approach to forming a common model
  - Given a black-box (BB) decision-making agent
  - Let x be an input to BB, with output x*
  - Assume that x* = argmin F(x), where F is a composite decision objective, local or global
  - Q: What is F?
  - Example: If a BB computes trajectories; F could be a composite objective combining distance and/or fuel and/or delays, etc.
  - Conjecture F(x) = $\sum w_i \, f_i(x)$ where $f_i(x)$ are known functions
  - Now use point ranking and compute the weights w_i

Objective Identification Process, cont.

- Ranking: Do until {}
  - Generate a set $\{x_i\}$ of trial solutions
  - Update BB ranking of the solutions by estimating the distance of the solutions from that obtained by the BB for the same input (a variety of criteria for BB's "satisfaction" with solutions are in analysis)
  - The ranked subset forms a set of vertices for search via optimization
- End do {
  - BB is "satisfied" with solution; we have a good representation of F or
  - Unable to approximate the BB criterion; continue generating solutions until run out or budget}

Algorithm, cont.

- To do: Compute a set of weights that will yield the same ordering as that given by the BB (results in a linear or quadratic program)
- Repeat until behavior of model and BB agree for test data and synthetic data
- If not, adjust basis functions or de-bug BB (structure), using synthetic data
- If a common set of weights is not detected, consider updating M or H model or both

Example: Objective Mismatch

- Consider the fundamental problem
  - Γ = set of feasible decisions (solutions $x^* \in \Gamma$)
  - $\Pi(n)$be the set of all Pareto points for this problem
  - Now add $f_{n+1}\ \Gamma \rightarrow \Re$, an additional objective
  - Gives rise to a new set of decisions $\Pi(n+1)$
- Mutual inclusions and intersection of $\Pi(n)$ and $\Pi(n+1)$ characterized under mild conditions
  - A large intersection signifies robustness with respect to objective formulation
  - Inclusion of original set in the expanded set signifies a greater number of solutions
  - Inclusion of the new set signified reduction in the number of solutions

Component: Predicting Phase Transitions

- Phase transition = shift from controllability to uncontrollability
- Essential requirement: anticipate in time to prevent
  - Timely anticipation obviates the need for “ethical” machine decisions
- Example of transition
  - Heterogeneity and density in the environment reduces system efficiency or makes conflict resolution intractable
  - Approach: complexity bounding
- Example of transition
  - System complexity is amenable to one agent’s decision making but not to another. E.g., machine agent can handle increased system complexity, but the human agent will not be able to take over should the need arise.
  - E.g., Problem gets too large; probabilistic reasoning required
  - Approach: context dependent warning, with potential complexity bounding via structural change

Concluding Remarks

- Touched on just a few of many questions in trustworthiness of machine decision making
- Risk will always be a deciding factor in complex adaptive systems
- Conjecture: for a single agent, traditional safety-based design and V&V likely covers a significant part of assuring safety
- For heterogeneous autonomous multi-agent systems, especially in a dense environment, justified trust is a complicated and open question
- Next steps: As the simulation environment matures, data in support of proposed method development will be assembled
- At the end of ATTRACTOR we are anticipating a general scheme for system design for trustworthy autonomous systems

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